

PREDICTING TYPE 2 DIABETES RISK

METHOD: Machine Learning & Explainable AI

GROUP: APEX
Capstone Project | 2026

THE CHALLENGE & SOLUTION

| THE CHALLENGE

Machine Learning models in healthcare lack transparency, making it difficult for clinicians to trust AI recommendations.

| THE SOLUTION

Machine Learning + Explainable AI (XAI)

- SHAP for global feature importance
- LIME for local, patient-level explanations

| THE GOAL

Build accurate disease risk prediction system with clinician-friendly explanations for each prediction.

KEY OBJECTIVES

DATA PREPROCESSING & EDA

- Clean and explore healthcare data
- Identify key risk factors for T2DM

MACHINE LEARNING MODEL DEVELOPMENT

- 4 algorithms: Logistic Regression, Decision Tree, Random Forest, XGBoost
- Compare and evaluate performance

EXPLAINABLE AI IMPLEMENTATION

- SHAP (global explanations) & LIME (local explanations)

INTERACTIVE DASHBOARD

- Streamlit web application for real-time predictions

DATASET & METHODOLOGY

| DATASET: BRFSS (CDC)

- 433,323 rows | 350 columns | 15 selected features
- Target: Diabetes
- Key features: BMI, Age, Sex, Health Status, Smoking, Physical Activity, Hypertension, Cholesterol

| METHODS OVERVIEW

1. Data Preprocessing: Scaling, encoding, outlier removal
2. EDA: Visualization, correlation analysis
3. ML Models: LR, DT, RF, XGBoost
4. XAI: SHAP (global), LIME (local)
5. Deployment: Streamlit Dashboard

PROJECT TIMELINE (12 Weeks)

Week	Activity	Deliverable
1-2	Data Collection & Preprocessing	Cleaned Dataset
3-4	EDA & Feature Engineering	EDA Report
5-6	Model Development & Tuning	Trained Models
7	Model Evaluation & Selection	Performance Report
8	XAI Implementation	SHAP + LIME Visuals
9-10	Dashboard Development	Streamlit App
11-12	Documentation & Presentation	Final Report + Slides

EXPECTED OUTCOMES

Category	Details
Recall	≥ 85% on test set
Performance	Outperforms Logistic Regression
XAI	SHAP & LIME visualizations generated
Dashboard	Fully functional Streamlit app
Deliverables	All items submitted on time

TECHNOLOGY STACK

Category	Tools / Technologies
Language	Python 3.8+
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn, XGBoost
Explainable AI	SHAP, LIME
Dashboard	Streamlit
Version Control	Git, GitHub

THANK YOU

| SUMMARY

Machine Learning + Explainable AI for Type 2 Diabetes Risk Prediction

| KEY BENEFITS

- Accurate predictions with transparent explanations
- Enhanced clinician trust and decision-making
- Interactive dashboard for real-time use

| FUTURE WORK

- Real-time clinical deployment
- Integration with EHR systems
- Expansion to other diseases

QUESTIONS?